(Please write your Enrollment number immediately) Enrollment No. ----------Hrs.

SUPPLEMENTARY MID TERM EXAMINATION

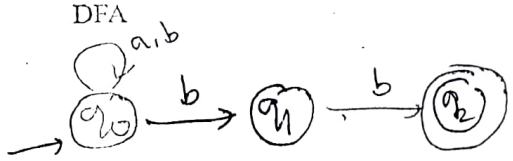
B.TECH PROGRAMMES (UNDER THE AEIS OF USICT)

IV Semester, March 2025

Paper Code: CIC-206 Theory of Computation

Time: $1\,{}^{1}/_{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No		Question	Max. Marks	CO(s)
1	(a)	Define regular expressions recursively	2	CO1
	(b)	Write regular expression for strings containing the pattern "*abac*" over $\sum = \{a, b, c\}$	2	CO1
	(c)	Write CFG for $L = \{a^n b^m \mid m > n\}$	2	CO2
	(d)	Define ambiguity in CFG. What is inherent ambiguity in CFL?	2	CO2
	(e)	Define Chomsky hierarchy of languages	2	CO1 & CO 2
2	(a)	Define DFA and NFA formally. Convert the following NFA to DFA 	5	CO1
	(b)	Show that $L = \{a^n b^n \mid n > 0\}$ is not regular using pumping lemma	5	CO1
3	(a)	Define a Pushdown Automaton. Write PDA for $L = \{wcw \mid w \in (a+b)^*\}$	5	CO2
	(b)	Define normal forms in CFG. Convert the following to CNF $S \rightarrow bA \mid aB, A \rightarrow bAA \mid aS \mid a, B \rightarrow aBB \mid bS \mid b$	5	CO2
4	(a)	Convert to GNF: $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$	5	CO2
	(b)	Prove that the following language is not context free a. $L = \{a^{n^2} b^n\}$	5	CO2

(Please write your Enrollment number immediately) Enrollment No. ----------Hrs.

SUPPLEMENTARY MID TERM EXAMINATION

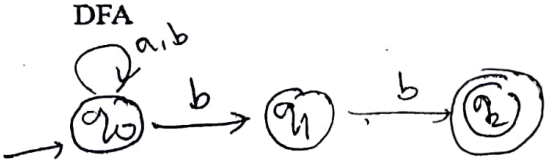
B.TECH PROGRAMMES (UNDER THE AEIS OF USICT)

IV Semester, March 2025

Paper Code: CIC-206 Theory of Computation

Time: $1\frac{1}{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No		Question	Max. Marks	CO(s)
1	(a)	Define regular expressions recursively	2	CO1
	(b)	Write regular expression for strings containing the pattern "*abac*" over $\sum = \{a, b, c\}$	2	CO1
	(c)	Write CFG for $L = \{a^n b^m \mid m > n\}$	2	CO2
	(d)	Define ambiguity in CFG. What is inherent ambiguity in CFL?	2	CO2
	(e)	Define Chomsky hierarchy of languages	2	CO1 & CO 2
2	(a)	Define DFA and NFA formally. Convert the following NFA to DFA 	5	CO1
	(b)	Show that $L = \{a^n b^n \mid n > 0\}$ is not regular using pumping lemma	5	CO1
3	(a)	Define a Pushdown Automaton. Write PDA for $L = \{wcw \mid w \in (a+b)^*\}$	5	CO2
	(b)	Define normal forms in CFG. Convert the following to CNF $S \rightarrow bA \mid aB, A \rightarrow bAA \mid aS \mid a, B \rightarrow aBB \mid bS \mid b$	5	CO2
4	(a)	Convert to GNF: $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$	5	CO2
	(b)	Prove that the following language is not context free a. $L = \{a^{n^2} b^n\}$	5	CO2

(Please write your Enrollment number immediately) Enrollment No. ----------Hrs.

SUPPLEMENTARY MID TERM EXAMINATION

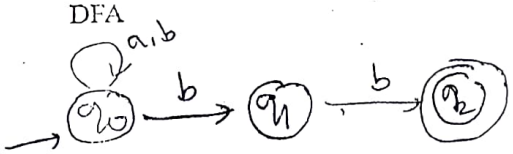
B.TECH PROGRAMMES (UNDER THE AEIS OF USICT)

IV Semester, March 2025

Paper Code: CIC-206 Theory of Computation

Time: $1\frac{1}{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No		Question	Max. Marks	CO(s)
1	(a)	Define regular expressions recursively	2	CO1
	(b)	Write regular expression for strings containing the pattern "*abac*" over $\sum = \{a, b, c\}$	2	CO1
	(c)	Write CFG for $L = \{a^n b^m \mid m > n\}$	2	CO2
	(d)	Define ambiguity in CFG. What is inherent ambiguity in CFL?	2	CO2
	(e)	Define Chomsky hierarchy of languages	2	CO1 & CO 2
2	(a)	Define DFA and NFA formally. Convert the following NFA to DFA 	5	CO1
	(b)	Show that $L = \{a^n b^n \mid n > 0\}$ is not regular using pumping lemma	5	CO1
3	(a)	Define a Pushdown Automaton. Write PDA for $L = \{wcw \mid w \in (a+b)^*\}$	5	CO2
	(b)	Define normal forms in CFG. Convert the following to CNF $S \rightarrow bA \mid aB, A \rightarrow bAA \mid aS \mid a, B \rightarrow aBB \mid bS \mid b$	5	CO2
4	(a)	Convert to GNF: $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$	5	CO2
	(b)	Prove that the following language is not context free a. $L = \{a^{n^2} b^n\}$	5	CO2

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SUPPLEMENTARY MID TERM EXAMINATION

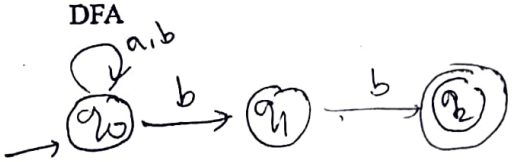
B.TECH PROGRAMMES (UNDER THE AEIS OF USICT)

IV Semester, March 2025

Paper Code: CIC-206 Theory of Computation

Time: $1\frac{1}{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No		Question	Max. Marks	CO(s)
1	(a)	Define regular expressions recursively	2	CO1
	(b)	Write regular expression for strings containing the pattern "*abac*" over $\sum = \{a, b, c\}$	2	CO1
	(c)	Write CFG for $L = \{a^n b^m \mid m > n\}$	2	CO2
	(d)	Define ambiguity in CFG. What is inherent ambiguity in CFL?	2	CO2
	(e)	Define Chomsky hierarchy of languages	2	CO1 & CO 2
2	(a)	Define DFA and NFA formally. Convert the following NFA to DFA 	5	CO1
	(b)	Show that $L = \{a^n b^n \mid n > 0\}$ is not regular using pumping lemma	5	CO1
3	(a)	Define a Pushdown Automaton. Write PDA for $L = \{wcw \mid w \in (a+b)^*\}$	5	CO2
	(b)	Define normal forms in CFG. Convert the following to CNF $S \rightarrow bA \mid aB, A \rightarrow bAA \mid aS \mid a, B \rightarrow aBB \mid bS \mid b$	5	CO2
4	(a)	Convert to GNF: $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$	5	CO2
	(b)	Prove that the following language is not context free a. $L = \{a^{n^2} b^n\}$	5	CO2

(Please write your Enrolment Number immediately) Enrolment No._____

MID TERM Examination

4th Semester, Mar 2025

Paper Code: CIC-206
Time: 1.5 Hrs.

Subject: Theory of Computation
Max Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining

Q1. Attempt any FIVE of the following. (2*5)

a) What is the difference between Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NDFA)? [CO1]

b) Define the term context free grammar with an example. [CO2]

c) Define instantaneous description of PDA. [CO2]

d) What are the disadvantages of unambiguous parse tree? Give an example. [CO2]

e) Describe the set of all Strings of 0's and 1's ending in 00 by Regular Expression. [CO1]

f) Write any five identity rules for regular expressions. [CO1]

Q2. a) Construct the DFA equivalent to NDFA whose transition table is given below (q0 – initial state, q4 – final state): (5) [CO1] Present

State	0	1	2
q0		q1, q4	q2	q2, q3
q1		-	q4	-
q2		q1, q3	-	q2, q4
q3		-	q4	-
q4		-	q1, q3	-

b) Prove the pumping lemma for regular languages with example. (5) [CO1]

Q3. a) Design a PDA for the language $\{a^n b^n, n>1\}$. (5) [CO2]

b)Construct a equivalent grammar G in CNF for the grammar G1 where G1=({S,A,B},{a,b},{S->bA/aB,A->bAA/aS/a,B->aBB/bS/b},S). (5) [CO2]

Q4. a) Explain Chomsky's Hierarchy of grammar in detail. (5 Marks) [CO1]

b) Prove that deterministic and non-deterministic PDA's are not equivalent.(5) [CO2]

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-206 **Subject: Theory of Computation**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Assume missing data, if any.

Q1 Answer the following questions: **(2.5x10=25)**

a) Write any five Identity Rules for Regular Expressions?

b) What is the difference between Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NDFA)?

c) What is the disadvantages of unambiguous parse tree. Give an example.

d) Define the term context free grammar with an example?

e) Define instantaneous description of a PDA

f) Construct the CFG for generating the language $L=\{a^n b^n /n>=1\}$.

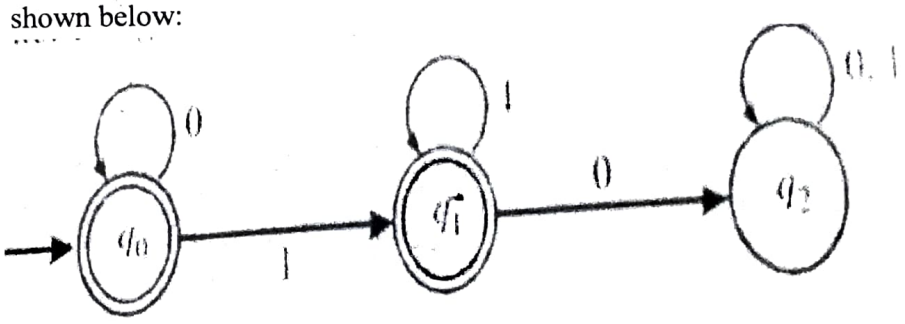
g) For the grammar S->A1B,A->0A | ε, B-> 0B | 1B| ε, give leftmost and rightmost derivations for the string 00101.

h) Prove graph coloring problem is NP complete.

i) Describe the Set of all Strings of 0's and 1's ending in 00 by Regular Expression.

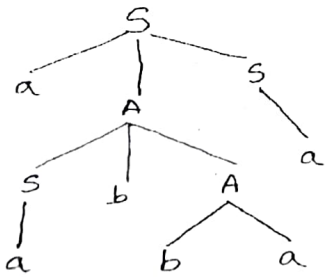
j) Construct a Regular Grammar G generating the regular expression 0*(0+1)?

Q2 a) Construct a Regular Expression for the FA (Using Method of Eliminate States) the shown below: **(6)**



b) Explain Pumping lemma for Context Free Languages. **(6.5)**

Q3 a) Find the left most and right most derivation corresponding to the tree. **(6)**



b) Construct a equivalent grammar G in CNF for the grammar G1 where G1 =({S,A,B},{a,b},{S->bA/aB,A->bAA/aS/a, B->aBB/bS/b},S) **(6.5)**

P.T.O.

P-1/2
CIC-206

(Please write your Exam Roll No.)

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETCS-206 **Subject: Theory of Computation**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Assume missing data, if any.

Q1 Differentiate between the following: **[5x5=25]**

a) Regular Language and Context Free language
b) Recursive and Recursive Enumerable language
c) Halting Problem & Un-decidable problem of Turing machine
d) Classify & Differentiate between different types of FA with output.
e) Dead state and In-accessible state

Q2 a) Explain closure properties of regular sets. Provide at least two closure properties of regular sets and justify their significance in formal language theory. **[4]**

b) Convert the following regular expression to DFA.
ab(a+b)*ab **[4]**

c) Explain the significance of the stack in a Pushdown Automaton with example. **[4.5]**

Q3 a) How the ambiguity of a grammar can be removed? Explain with suitable example. **[6]**

b) How to Convert any CFG to GNF? Explain with suitable example. **[6.5]**

Q4 a) What are Normal Forms in grammar? Explain its importance with example. **[4]**

b) Explain two types of acceptance in pushdown automata for a^nb^n. **[4.5]**

c) Define Pushdown Automata and explain its components. **[4]**

Q5 Write a short note on following

a) Kleen's closure & Positive Closure **[4]**
b) Undecidability **[4.5]**
c) SPACE and PSPACE **[4]**

Q6 a) Write the steps and convert the regular grammar into Finite Automata for **[6.5]**
$S \rightarrow 0A/1B0/1$ $A \rightarrow 0S/1B/1$ $B \rightarrow 0A/S$

b) Construct a TM for a^nb^n. **[6]**

Q7 a) Design a Turing machine for string of palindrome (Same machine for even and odd length). **[7.5]**

b) Write a short note on Computability and Intractability. **[5]**

P.T.O.

P 1/2
ETCS-206

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

IV Semester, April, 2024

Paper Code: CIC-206 **Subject: Theory of Computation**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1			
a.	Define Regular expression.	2.5	CO1
b.	Prove that Regular languages are closed under intersection.	2.5	CO1
c.	Define Ambiguity in CFL with the help of an example.	2.5	CO2
d.	State and prove pumping Lemma for Context Free Grammar.	2.5	CO2
2			
a.	Design NDFA for a language over input alphabet {0,1} accepting strings in which number of 1s are divisible by either three or five.	5	CO1
b.	Design a Push Down Automata for L(G)={$a^n b^n$, n>=0}	5	CO2
3			
a.	Convert the following NFA to DFA State 0 1 q0 q0, q1 q0 q1 q1 q2 q2 q1, q2 q0 q0 is the initial state and q2 is the only final state.	5	CO1
b	Define LL(K) parsing in detail with the help of an example.	5	CO2
4			
a.	Find the regular expression equivalent to following FA State a b q1 q1 q2 q2 q3 q2 q3 q2 q1 q1 is the initial state and q1 is the only final state	5	CO1
b.	Define Normal Forms for Grammar. Convert the following CFG in to Chomsky normal form S→aAb| bBa| Ɛ A→abc|aAB B→cba	5	CO2

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END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] July 2023

Paper Code- ETCS 206	**Subject: Theory of Computation**
Time: 3 Hours	**Maximum Marks:75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt all questions: **(5x5=25)**

(a) State the differnces between a NFA and DFA?
(b) Differentiate between Moore and Melay Machines?
(c) Find a regular expression corresponding to the language of all strings over the alphabet { a, b } that contain exactly two a's.
(d) State Church's hypothesis about compuitability of a machine
(e) Prove that the graph coloring problem is NP-complete

UNIT-I

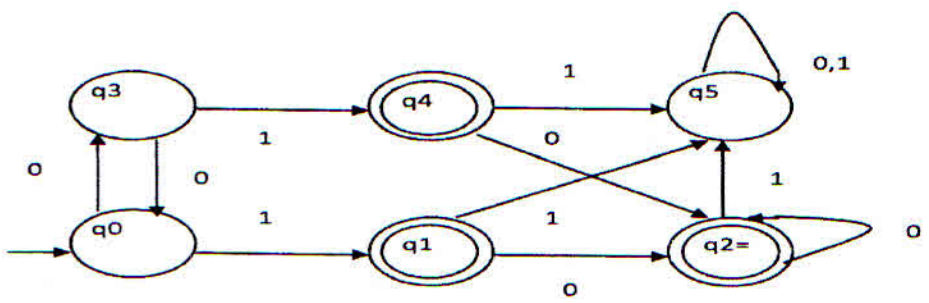
Q2 a) Let Σ = {a, b} and let L = { w $\in \Sigma^*$ | w $\neq \varepsilon$ and the first and last character of w are the same }. Design aDFA for L **(6)**

b) For the following NFA. Convert it into its equivalent DFAwhere Q0 is the initial state and Q2 is the final state. **(6.5)**

	a	b
Q0	{Q0}	{Q0,Q1}
Q1	€	{Q2}
Q2	€	€

Q3 a) Prove the pumping lemma for regular lanuages with examples. **(6)**

b) Minimize the following DFA over alphates as 0,1 **(6.5)**



UNIT-II

Q4 a) What are inherently ambiguous languages. Check whether the folllowing Grammar is Ambiguous or not-S->A/B, A->aAb /ab, B->abB/ € **(6)**

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P-1/2
ETCS-206